

Strawberry modulates LDL oxidation and postprandial lipemia in response to high-fat meal in overweight hyperlipidemic men and women.

BACKGROUND: Elevated levels of lipids, such as total cholesterol (TC), low-density lipoprotein cholesterol (LDL), and triglycerides (TG), are widely recognized as risk factors for cardiovascular disease (CVD). Oxidized LDL (OxLDL) is an emerging risk factor considered relevant in oxidative stress and endothelial dysfunction, which is implicated in the progression of CVD. Consumption of a diet rich in polyphenols may be cardioprotective through its impact on oxidative stress and protecting LDL from oxidation. **OBJECTIVES:** This study was designed to test the ability of strawberry phenolic compounds to mitigate the postprandial effects of a high-fat meal on OxLDL as well as investigate the effects of phenolic compounds on lipid metabolism. **METHODS:** Twenty-four hyperlipidemic men and women (14 women, 10 men; mean age 50.9 +/- SD 15 years) were recruited to participate in this randomized, single-blind, placebo-controlled, 12-wk crossover trial. After a 10-day run-in period, subjects consumed either an active strawberry beverage (Str; containing 10 g freeze-dried fruit) or a placebo (Pbo) beverage matched in energy and macronutrient composition for 6 weeks. Twice before randomization and once at the 6-week crossover point, subjects received either Str or Pbo with a high-fat challenge meal (HFM). TC, LDL, high-density lipoprotein cholesterol, TG, and OxLDL were measured at defined intervals for 6 h before and after HFM challenge. Fasting concentrations of blood variables at 0, 6, and 12 weeks were compared to assess chronic intake of Str or Pbo. **RESULTS:** After the HFM during the run-in period, TG and OxLDL were lower after Str than Pbo ($p = 0.005$, $p = 0.01$, and $p = 0.0008$, respectively). HFM responses after 6 weeks of Str versus Pbo resulted in decreased lipid levels and a sex by treatment interaction for OxLDL ($p < 0.0001$, and $p = 0.0002$). **CONCLUSION:** The present results support a role for strawberry in mitigating fed-state oxidative stressors that may contribute to atherogenesis.